

SHIV NADAR
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Use-Case Document

for

Video Rental Store Management System (VRSMS)

Version 1.2

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Reference Documents:

VRSMS SRS v1.2, Group 23 DFDs (Context & Level 1)

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1. Problem Statement and Context

The Video Rental Store Management System (VRSMS) is a web-based management platform supporting a physical video rental store. Primary functions include membership management, rental issue/return (enforcing a 10-day loan limit), inventory configuration, scheduled maintenance tasks (auto-cancellations, rate halving), and administrative reporting. Following the v1.2 architecture update, the system integrates an OTP-based remote authentication mechanism, dynamic wishlist features, and enforces strict overdue fines and auto-cancellation rules.

2. Traceability Summary

Use Case ID	Title	SRS / DFD Ref.	Primary Actor
UC-REG-01	Register Member	0.1 Manage Mem.	Store Clerk
UC-LOGIN-01	Remote Login (OTP)	0.5 Member Svcs.	Member
UC-VERIFY-01	Verify OTP	0.5 Member Svcs.	System (Auth)
UC-CANCEL-01	Cancel Membership	0.1 Manage Mem.	Store Clerk
UC-AUTO-01	Auto-Cancel Membership (scheduled)	0.1 Manage Mem.	System (Scheduler)
UC-ISSUE-01	Issue Rental	0.3 Rentals & Ret.	Store Clerk
UC-RETURN-01	Process Return / Calculate Fine	0.3 Rentals & Ret.	Store Clerk
UC-LOSS-01	Report Loss or Damage	0.3 Rentals & Ret.	Store Clerk
UC-SEARCH-01	Search Catalog (incl. Wishlist)	0.5 Member Svcs.	Member / Clerk
UC-ADD-01	Add New Item	0.2 Manage Inv.	Store Manager
UC-RATE-01	Update Rental Rate	0.2 Manage Inv.	Store Manager
UC-SELL-01	Dispose Unsold Item	0.2 Manage Inv.	Store Manager
UC-CONFIG-01	Configure System Parameters	0.4 Admin & Rep.	Store Manager
UC-REPORT-01	Generate Reports	0.4 Admin & Rep.	Store Manager

3. Use Case Descriptions

3.1. UC-REG-01 Register Member

ID	UC-REG-01
Title	Register Member
Description	The Store Clerk registers a new customer by capturing demographic details, verifying their phone number via OTP, capturing a webcam photo, and collecting the security deposit.
Primary Actor	Store Clerk
Secondary Actor	SMS Gateway, Receipt Printer
Preconditions	The Clerk is logged into the store terminal, and the customer has a valid phone number and the required deposit.
Postconditions	A new Member record is created in the database with an 'Active' status and the registration receipt is printed.
Dependency	«include» UC-VERIFY-01
Generalization	None
Main Success Scenario	1. Clerk initiates "New Member Registration" and enters the customer's phone number. 2. System requests OTP generation (invokes UC-VERIFY-01). 3. SMS Gateway sends a 6-digit OTP to the customer's phone. 4. Clerk enters the OTP provided by the customer; system verifies it successfully. 5. System prompts the Clerk to capture the user photo via webcam (HW-1). 6. Clerk submits the demographic details and the deposit payment reference. 7. System creates the account and generates a Registration Receipt.
Extensions or Alternate Flow	3a. SMS delivery fails: System displays delivery failure. Clerk can request a new OTP or abort. 4a. OTP invalid: System rejects entry. After 3 failed attempts, registration is aborted. 6a. Deposit payment fails: Registration is aborted; member record is not saved.
Frequency of Use	High (daily as new customers visit the store)
Status	Ready for Development
Owner	Group 23
Priority	High

3.2. UC-LOGIN-01 Remote Login (OTP)

ID	UC-LOGIN-01
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Title	Remote Login (OTP)
Description	A registered Member requests remote login to the web portal. The system sends an OTP to their registered phone number and authenticates them upon successful entry.
Primary Actor	Member
Secondary Actor	SMS Gateway
Preconditions	The Member is already registered and their account status is 'Active'.
Postconditions	The Member is authenticated and a secure session (HTTPS) is established.
Dependency	«include» UC-VERIFY-01
Generalization	None
Main Success Scenario	1. Member accesses the web portal and enters their registered phone number. 2. System verifies the phone number exists and triggers UC-VERIFY-01. 3. Member receives the OTP and enters it into the portal. 4. System validates the OTP and grants access to the Member Dashboard.
Extensions or Alternate Flow	2a. Phone number not found: System displays an error prompting the user to register in-store. 2b. Account Cancelled: System denies login and displays a contact support message.
Frequency of Use	High
Status	Ready for Development
Owner	Group 23
Priority	High

3.3. UC-VERIFY-01 Verify OTP

ID	UC-VERIFY-01
Title	Verify OTP
Description	The system generates a time-sensitive passcode and validates it against the user's input to authenticate identity without passwords.
Primary Actor	System (Authentication Module)
Secondary Actor	SMS Gateway (SI-1)
Preconditions	An authentication or registration flow has requested verification.
Postconditions	The system returns a success or failure boolean to the calling use case.
Dependency	None (This use case is included by others)

Generalization	None
Main Success Scenario	1. System generates a cryptographically secure 6-digit OTP. 2. System dispatches the OTP via the SMS Gateway API to the target phone number. 3. User inputs the OTP within the validity window. 4. System confirms the match and returns a success status.
Extensions or Alternate Flow	2a. Gateway Timeout: System logs API failure and alerts the user to try again later. 3a. OTP Expired: System rejects the input and prompts the user to request a new OTP.
Frequency of Use	High (triggered by all logins and registrations)
Status	Ready for Development
Owner	Group 23
Priority	High

3.4. UC-CANCEL-01 Cancel Membership

ID	UC-CANCEL-01
Title	Cancel Membership
Description	Clerk initiates membership cancellation; system verifies no outstanding loans or fines, processes deposit refund if applicable, and marks the member's status as Cancelled.
Primary Actor	Store Clerk
Secondary Actor	Database
Preconditions	Member exists and account is active.
Postconditions	Member status updated to Cancelled; deposit refund processed or flagged.
Dependency	None
Generalization	None
Main Success Scenario	1. Clerk searches for and selects the member record. 2. System checks DB for outstanding loans/fines. 3. If no dues/loans, system processes deposit refund and sets member status to Cancelled. 4. System prints a cancellation receipt and logs the transaction.
Extensions or Alternate Flow	2a. Outstanding loans exist: Deny cancellation until items are returned. 2b. Outstanding fines: Require fine payment before cancellation can proceed.
Frequency of Use	Low
Status	Ready for Development

Owner	Group 23
Priority	Medium

3.5. UC-AUTO-01 **Auto-Cancel Membership (scheduled)**

ID	UC-AUTO-01
Title	Auto-Cancel Membership (scheduled)
Description	Scheduled job automatically identifies members meeting configured criteria for cancellation (e.g., severe overdue fines over the Cancellation Threshold) and triggers cancellation processing.
Primary Actor	System (Scheduler)
Secondary Actor	Database
Preconditions	System configuration for auto-cancel threshold is set.
Postconditions	Eligible members processed for cancellation; audit log created.
Dependency	«extend» UC-CANCEL-01
Generalization	None
Main Success Scenario	1. Scheduler triggers the auto-cancel job per configured frequency (e.g., nightly). 2. System queries DB for members whose fines exceed the Cancellation Threshold. 3. System sets member status to 'Cancelled' and flags deposit to cover fines. 4. System logs action for the Manager's Auto-Cancellation Alert report.
Extensions or Alternate Flow	2a. No members exceed the threshold: System logs successful job completion and terminates.
Frequency of Use	Periodic (Daily batch process)
Status	Ready for Development
Owner	Group 23
Priority	Low

3.6. UC-ISSUE-01 **Issue Rental**

ID	UC-ISSUE-01
Title	Issue Rental
Description	The Store Clerk issues an available item to a Member, enforcing the maximum 10-day loan duration rule and generating a receipt.
Primary Actor	Store Clerk
Secondary Actor	Receipt Printer (HW-2)

Preconditions	Member is physically present, account is active, and the item status is 'Available'.
Postconditions	A Loan Record is created, item status updates to 'On Loan', and expected return date is set.
Dependency	None
Generalization	None
Main Success Scenario	1. Clerk inputs the Member ID and the Item ID into the POS terminal. 2. System verifies Member has no unresolved overdue fines or auto-cancellation alerts. 3. System verifies item is available. 4. System calculates the expected return date (Issue Date + 10 days max limit). 5. System creates the Loan Record and updates Inventory DB. 6. System prints a physical transaction slip.
Extensions or Alternate Flow	2a. Member has unresolved fines: System blocks the issue process until fines are cleared. 3a. Item not available: System displays error and aborts transaction.
Frequency of Use	High
Status	Ready for Development
Owner	Group 23
Priority	High

3.7. UC-RETURN-01 Process Return / Calculate Fine

ID	UC-RETURN-01
Title	Process Return / Calculate Fine
Description	The Clerk processes a returned item. The system calculates any overdue fines based on the Config Parameters and updates the database.
Primary Actor	Store Clerk
Secondary Actor	Database
Preconditions	A valid Loan Record exists for the returned Item ID.
Postconditions	The Loan Record is closed, fine amounts are recorded, and Item Status reverts to 'Available'.
Dependency	None
Generalization	None

Main Success Scenario	1. Clerk inputs the returned Item ID. 2. System retrieves the active Loan Record and expected return date. 3. System calculates overdue days (if returned past 10-day limit). 4. System applies the Overdue Fine Amount multiplier (set by Manager). 5. System displays total dues to the Clerk. 6. Clerk confirms receipt of item (and payment of fines, if applicable). 7. System closes the loan and marks the item Available.
Extensions or Alternate Flow	6a. Member refuses to pay fine: System marks item as Available but adds fine amount to the Member's Current Dues, blocking future rentals. 6b. Item is physically damaged: System branches to UC-LOSS-01.
Frequency of Use	High
Status	Ready for Development
Owner	Group 23
Priority	High

3.8. UC-LOSS-01 Report Loss or Damage

ID	UC-LOSS-01
Title	Report Loss or Damage
Description	Clerk records lost or damaged items, computes applicable charges, removes the item from inventory, and updates the member's account accordingly.
Primary Actor	Store Clerk
Secondary Actor	Database
Preconditions	There is a loan record for the item, or damage is observed during return.
Postconditions	Item marked as Damaged/Removed; member billed for replacement; inventory updated.
Dependency	Can be extended from UC-RETURN-01
Generalization	None
Main Success Scenario	1. Clerk records incident (lost/damaged) and attaches notes. 2. System fetches item purchase price and computes the replacement charge. 3. Clerk collects payment or records the outstanding charge on member account. 4. System marks the item as Removed/Damaged in Inventory DB and logs the transaction.

Extensions or Alternate Flow	3a. Member disputes charge: Create dispute ticket and hold final removal until manager resolution.
Frequency of Use	Low
Status	Ready for Development
Owner	Group 23
Priority	Medium

3.9. UC-SEARCH-01 Search Catalog and History

ID	UC-SEARCH-01
Title	Search Catalog and History
Description	Allows Members or Clerks to search the store's inventory by keywords and apply filters to view item availability and details. Members can add items to their Wishlist.
Primary Actor	Member / Store Clerk
Secondary Actor	Database
Preconditions	The user is accessing the web portal or POS system.
Postconditions	Search results are displayed within 2 seconds.
Dependency	None
Generalization	None
Main Success Scenario	1. User enters keywords (Title, Genre) or selects filters (Category, Format). 2. System queries the central database. 3. System retrieves and paginates matching Item Records. 4. System displays the list including Title, Format, Daily Rate, and Status (Available/On Loan).
Extensions or Alternate Flow	2a. No matching items: System displays "No results found" and suggests altering filters. 4a. Authenticated Member flow (Wishlist): System allows Member to click "Add to Wishlist" next to catalog items, modifying their personal profile preferences.
Frequency of Use	Very High
Status	Ready for Development
Owner	Group 23
Priority	High

3.10. UC-ADD-01 Add New Item

ID	UC-ADD-01
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Title	Add New Item
Description	Manager or Clerk enters newly procured item details (title, format, purchase price, daily rental charge, procurement date) into the Inventory DB.
Primary Actor	Store Manager
Secondary Actor	Database
Preconditions	Manager authenticated.
Postconditions	New item record created with status = Available.
Dependency	None
Generalization	None
Main Success Scenario	1. Manager selects “Add New Item.” 2. System prompts for required metadata (Title, Format, Purchase Price). 3. Manager enters details and confirms. 4. System validates and adds the item to Inventory DB. 5. Item becomes searchable in the catalog.
Extensions or Alternate Flow	3a. Missing mandatory data: System asks user to fill required fields before saving.
Frequency of Use	Moderate
Status	Ready for Development
Owner	Group 23
Priority	Medium

3.11. UC-RATE-01 Update Rental Rate (halve after 1 year)

ID	UC-RATE-01
Title	Update Rental Rate (halve after 1 year)
Description	System automatically halves daily rental charge for items older than one year according to business rules; Manager can also manually trigger or override this change.
Primary Actor	Store Manager / System Scheduler
Secondary Actor	Database
Preconditions	Item purchase date present in DB.
Postconditions	Rental rates updated and change history logged.
Dependency	None
Generalization	None

Main Success Scenario	1. Scheduler runs the periodic rate-halving job (or Manager triggers manually). 2. System identifies items with purchase date exactly one year older than current date. 3. System halves the daily rental charge and updates Inventory DB. 4. System logs the change.
Extensions or Alternate Flow	2a. Administrator configures exceptions: Certain items excluded from auto-halving.
Frequency of Use	Periodic (Daily check)
Status	Ready for Development
Owner	Group 23
Priority	Low

3.12. UC-SELL-01 Dispose Unsold Item (Dead Stock Sale)

ID	UC-SELL-01
Title	Dispose Unsold Item (Dead Stock Sale)
Description	System identifies items not lent out for over a year, computes a dead-stock sale price, and records the manual sale and removal from active inventory by the Manager.
Primary Actor	Store Manager
Secondary Actor	Database
Preconditions	Manager authenticated.
Postconditions	Item marked Sold/Removed; sale recorded; inventory adjusted.
Dependency	None
Generalization	None
Main Success Scenario	1. Manager queries items which have never been lent or not lent for a year. 2. System flags candidates and calculates a suggested sale price (10% of purchase price). 3. Manager approves sale to a third-party vendor. 4. System records transaction and marks item removed from active inventory.
Extensions or Alternate Flow	2a. Manager chooses a different price: System accepts manual override and records final price.
Frequency of Use	Low (Annual or Bi-annual)
Status	Ready for Development
Owner	Group 23
Priority	Low

3.13. UC-CONFIG-01 Configure System Parameters

ID	UC-CONFIG-01
Title	Configure System Parameters
Description	The Store Manager sets global numerical variables (Max Loan Duration, Overdue Fine Amount, Cancellation Threshold) affecting business logic.
Primary Actor	Store Manager
Secondary Actor	Database
Preconditions	The Manager is authenticated on the Admin Dashboard.
Postconditions	New parameters are saved and immediately applied to subsequent transactions.
Dependency	None
Generalization	None
Main Success Scenario	1. Manager navigates to the Configuration tab. 2. System displays current values for Loan Duration, Fines, and Thresholds. 3. Manager overwrites specific numerical fields and clicks Save. 4. System validates the inputs (e.g., ensuring no negative integers). 5. System commits changes to the database and displays a success banner.
Extensions or Alternate Flow	4a. Invalid input type: System highlights the field in red and prevents saving until corrected.
Frequency of Use	Low (only when store policies change)
Status	Ready for Development
Owner	Group 23
Priority	Medium

3.14. UC-REPORT-01 Generate Reports

ID	UC-REPORT-01
Title	Generate Reports (Profit/Loss, Dead Stock)
Description	Manager generates operational and financial reports (profit/loss account, dead-stock reports, transaction history, auto-cancellation alerts).
Primary Actor	Store Manager
Secondary Actor	Database
Preconditions	Required transaction and inventory data present.
Postconditions	Report generated and available for viewing/download.
Dependency	None

Generalization	None
Main Success Scenario	1. Manager selects report type and date range (profit/loss, dead stock). 2. System aggregates data from DB and renders report. 3. Manager views the report on-screen.
Extensions or Alternate Flow	2a. Insufficient data for range: System informs manager and offers partial results or prompts for new date range.
Frequency of Use	Moderate (periodic financial checks)
Status	Ready for Development
Owner	Group 23
Priority	Medium